ANALYSTLAB AFRICA

Data Quality Assessment

Hospital Operations, Cost & Patient Flow Intelligence | Data Analyst | Prepared by Kukoyi Zainab Omotola

1. Purpose and approach

This assessment records the condition of the supplied healthcare data before it was used in the dashboard. Each file was profiled for record and column counts, missing values, duplicates, data types, date validity, key relationships and unusual values. No duplicates were removed and no missing values were replaced; each issue was first understood and then given a documented decision.

2. Files inspected

File	Records	Columns	Role in the analysis
encounters.csv	27,891	14	Main fact table: volumes, classes, duration, costs, payer coverage
patients.csv	974	20	Patient-level context only; identifying columns removed
procedures.csv	47,701	9	Procedure frequency and base cost
payers.csv	10	7	Payer names for payer analysis
organizations.csv	1	8	Context only (one organization)
data_dictionary.csv	65	3	Field definitions; used for reference, not loaded

3. Checks that passed

- No duplicate rows, no duplicate encounter Ids and no duplicate procedure rows.
- No blank values in key fields (Id, PATIENT, PAYER, ORGANIZATION, ENCOUNTERCLASS in encounters; ENCOUNTER, PATIENT, CODE in procedures).
- No orphan keys: every encounter links to a patient, payer and organization, and every procedure links to an encounter. Every procedure's patient matches the patient on its encounter.
- START and STOP are present and valid in every encounter, none ends before it starts, and none has zero length.
- No negative costs, and payer coverage never exceeds the total claim cost.
- Encounter class has six consistent values (ambulatory, outpatient, urgent care, emergency, wellness, inpatient).

4. Issues found and decisions

#	Table	Finding	Impact	Decision
1	encounters: PAYER_COVERAGE	0 in 13,586 encounters (48.7%). 8,807 are NO_INSURANCE; 4,779 are with named insurers. Five private insurers record 0.0% to 0.2% of claim cost as covered.	High	Kept and reported as recorded. Meaning (uninsured, unrecorded or unpaid) is not in the data, so no cause is assumed.
2	encounters: duration	Median duration is 15 minutes. 75 encounters exceed 24 hours, of which 13 exceed 30 days and 4 exceed one year (maximum 44,930 hours, about 5.1 years; all ambulatory or outpatient).	Medium	Kept. The over-24-hour KPI is reported with this caveat, because the extreme values look like long-running care episodes rather than stays.

#	Table	Finding	Impact	Decision
3	procedures: START, STOP	In 24,388 procedures (51.1%) the start time falls outside the time window of their encounter (2,060 before it starts, 22,328 after it ends).	Medium	Procedure timestamps are not used for timing. Procedures are placed in time using their encounter's start date.
4	encounters without procedures	13,221 encounters (47.4%) have no procedure record.	Medium	Valid, since not every encounter has a procedure. Procedure analysis is described as covering part of the activity.
5	encounters: TOTAL_CLAIM_COST	0 in 89 encounters, so claim cost is below base cost in the same 89.	Low	Kept. Their effect on averages is small and noted.
6	encounters / procedures: REASONCODE, REASONDESCRIPTION	Blank in 19,541 encounters (70.1%) and 36,945 procedures (77.5%).	Low	Not used. Columns dropped in Power Query.
7	procedures: CODE and DESCRIPTION	Four codes appear under more than one description (for example depression screening has two wordings). 163 descriptions share 157 codes.	Low	Procedures are counted by description. This can split one procedure across rows and is noted as a limitation.
8	procedures: BASE_COST	57 records are under $10 and the highest is $289,531. Procedure base costs sum to about $105.5M, more than the total claim cost of $101.5M.	Low	All kept. Base cost is treated as a different measure from claim cost and the two are never added or compared directly.
9	data dictionary	Middle and FIPS County Code are listed for patients but are not in patients.csv.	Low	Noted. Only fields confirmed in the files were used.
10	patients: blanks and identifiers	DEATHDATE blank for 820 (154 deceased), ZIP for 142, SUFFIX for 953, MAIDEN for 588, MARITAL for 1. The file also holds names, addresses and other identifiers.	Low (blanks), High (privacy)	Blanks not used. All identifying columns removed in Power Query; only Id, birth date and gender were loaded.
11	encounters: START, STOP	Stored as text with a UTC marker (for example 2011-01-02T09:26:36Z).	Low	Loaded as date/time with time zone so dates are not shifted to local time.
12	all cost fields	The currency is not stated.	Low	Costs are shown with a dollar sign for presentation, and the missing currency is listed as a limitation.
13	payers: NO_INSURANCE	Address, city, state, zip and phone are blank.	None	Expected, because it is not a real insurer.

5. Transformations performed and why

Table	Transformation	Reason
All	Promoted headers and set data types	Files load with untyped columns
encounters	Kept 11 of 14 columns (dropped CODE, REASONCODE, REASONDESCRIPTION)	Dropped fields are not used and are mostly blank
encounters	Converted START and STOP to date/time with time zone	Timestamps are UTC text
encounters	Added Start Date and Duration Hours	Needed for the date relationship, the trend and the over-24-hours KPI

Table	Transformation	Reason
procedures	Kept 7 of 9 columns (dropped REASONCODE, REASONDESCRIPTION)	Dropped fields are mostly blank and not used
patients	Kept 3 of 20 columns (Id, BIRTHDATE, GENDER)	Removes names, addresses and other identifiers for privacy
payers, organizations	Kept Id and NAME only	Other fields are not needed
Model	Added a date table (2011 to 2022) and the relationships encounters to patients, payers, organizations and date, and procedures to encounters	Needed for filtering and time analysis

No rows were removed, and no values were replaced or imputed.

6. Fitness for purpose

The data is sound enough for the operational and financial questions in this assignment: structure and keys are clean, and the main limits are about what the fields mean rather than about errors. The findings should be read with these limits in mind:

- One hospital only, and 2022 contains data only to 5 February.
- February 2014 has 1,179 encounters, against about 200 in a typical month. The data does not explain it.
- Payer coverage of zero cannot be interpreted without billing and payment data.
- Procedure timing, procedure cost and the over-24-hour measure each have the caveats listed in section 4.
- No clinical outcomes are included, so no clinical claims are made.

7. Suggested data improvements

- Add payment, write-off or patient-responsibility fields so zero coverage can be interpreted.
- Align procedure timestamps with encounter times, or document why they differ.
- Use one description per procedure code, and state the currency and the meaning of base cost versus claim cost in the data dictionary.
- Explain the February 2014 volume and the very long encounter durations.